2025 International Conference on Quantum Photonics, Artificial Intelligence, and Networking (QPAIN)
31 July - 2 August 2025, Rangpur, Bangladesh.

IMU-IoT Integrated Foot Exoskeleton for Real-time Gait Analysis and Monitoring.

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***Abstract*—Post diabetic Foot Drop syndrome significantly impacts mobility and quality of life. To address this issue, a low-cost IMU-based exoskeleton was developed focusing on ankle movement and integrating IoT for real-time gait analysis and foot health monitoring. The system incorporated an MPU6050 sensor for motion tracking and a DHT22 sensor for temperature and humidity sensing, both of which are managed by an Arduino Uno and an ESP32 microcontroller. Sensor data was transmitted wirelessly to a cloud-based open-source platform (ThingSpeakTM). Real-time data logging and downloadable CSV format sensor data were integrated. A tendon-like pulling mechanism, actuated by a servo motor, was implemented to replicate the movement of the ankle. This prototype was constructed using low-cost materials, such as spare aluminium parts and 3D-printed TPU. Future improvements include durable structural materials, higher-torque motors, a personalized real-time data dashboard, and compact circuitry. Despite these limitations, this project demonstrates the feasibility of a smart, wearable, and affordable exoskeleton.**

Keywords— exoskeleton, IMU sensor, foot drop, IOT, rehabilitation, telemedicine, gait analysis

I. INTRODUCTION

Rehabilitation has become a burning research sector in recent times. Many people in our country have paralysis or partial paralysis originating from diabetes. This project aims to implement a low-cost smart exoskeleton to help patients suffering from a specific muscle disease, Foot drop. Usually, Exoskeletons are expensive, and considering our country's socio-economic situation, a low-cost rehabilitation solution is in dire need. Foot drop is a term used to describe the weakening of the muscles responsible for lifting the front part of the foot. Foot drop refers explicitly to the reduced strength in the dorsiflexor muscles [1].

The leading cause of foot drop is Peripheral Neurology, such as Neuropathy caused by diabetes. Treatment for foot drop includes assistive devices tailored to the specific underlying cause. The most commonly used orthosis for foot drop is the ankle-foot orthosis (AFO). Typically made of plastic, it holds the ankle at a 90-degree angle [2]. However, conventional AFOs do not always yield satisfactory results. In general, there are three types of ankle-foot orthotic (AFO) devices: passive, semiactive, and active devices. Active Ankle-Foot Orthoses (AFOs) are equipped with an onboard power source, control system, sensors, and actuators. These designs can enhance gait, prevent foot drop, and offer benefits in terms of safety, cost, and compactness. The motion control capabilities of passive Ankle-Foot Orthoses (AFOs) are constrained by their passive components. Active Ankle-Foot Orthoses (AFOs) can adapt to the walking environment and respond appropriately. The control system typically includes elements such as a force sensor, an angle-measuring sensor, an accelerometer, and a microprocessor [3].

Exoskeletons, controlled by sensors and AI, provide significant support for rehabilitation. However, challenges such as device security and continuous control remain. Proposed models currently employ big data analytics and the Internet of Things (IoT) to enhance smart exoskeletons, thereby improving the efficiency and usability of rehabilitation devices [4].

II. LITERATURE REVIEW

Shefa et al. [5] presents a custom AFO made from 3D-printed polypropylene, instrumented with a small inertial measurement unit (IMU) and a surface EMG sensor. The IMU captures gait acceleration and orientation. Data is transmitted via Wi-Fi to a cloud-based backend where machine learning models analyze the data. Callihan et al. [6] used MATLAB for gait computation, collecting pressure sensor data from the smart shoe, and used MATLAB R2019b's classification tools to predict discrete ankle joint angles. Long et al. [7] used ThingSpeakTM by uploading one-hour IMU recordings (100 Hz tri-axial recordings from AX3 sensors) and running MATLAB gait computation in the cloud. Casasola et al. [8] designed a smart splint that monitors skin temperature and humidity under a cast. The device used a DHT22 humidity and temperature sensor at the skin-splint interface. The microcontroller digitized these signals so that they can be relayed wirelessly.

In the field of rehabilitation, the most significant barriers are affordability and availability. As Bangladesh is a developing country, Exoskeletons are not readily available for their high price. As diabetes is a common disease in Bangladesh and Foot drop frequently leads to complications where simple loss of mobility of the foot eventually leads to severance of the leg via surgery, this project is in dire need.

This project takes a significant step toward solving the accessibility problem of modern exoskeletons by focusing on low-cost design and IoT integration. Its primary goal is to make smart rehabilitation technology available to the general population, especially in rural areas of Bangladesh. By utilizing affordable materials, this prototype significantly reduces costs. What sets this project apart is its ability to wirelessly monitor foot health, sending real-time data to the cloud, where it can be remotely accessed and analyzed by physicians. Ultimately, this project shows it is possible to create a smart, wearable, and affordable exoskeleton that meets the practical needs of patients in low-resource settings.

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III. METHODOLOGY

The system utilizes a DHT22 sensor to measure temperature and humidity, enhancing user comfort. An adjustable servo motor dynamically supports ankle movement in response to the MPU6050 sensor's Y-axis movement. The collected data is transmitted for analysis and visualization, enabling healthcare providers to develop personalized treatment plans.

A. System Architecture

The system design was carefully done to ensure proper movement of the ankle model. It utilized the y-axis rotation of the MPU6050 sensor. Again, a threshold was set to maintain a degree of rotation. A block diagram (Fig. 1) illustrates the interconnection between the main components, including Arduino Uno, motors, sensors, and a power supply.

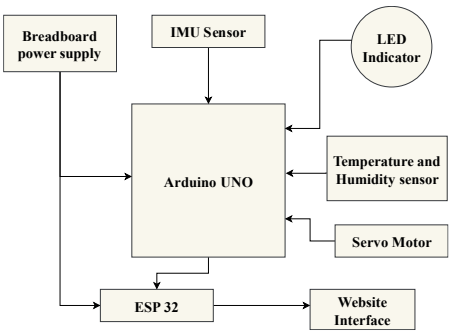


Fig. 1. Block diagram of the IMU based exoskeleton using IOT for real-time gait analysis and foot health monitoring.

B. Componets Description

TABLE I. COMPONENTS USED TO IMPLEMENT THE SYSTEM.

Component name	Description	Quantity
Arduino UNO R3	Microcontroller	01
DHT22 Original AM2302	Digital Temperature-Humidity sensor	01
FSR	Pressure sensor	02
Li-Battery	3.7V 3000mWh	02
ESP 32 WROOM 32	Microcontroller	01
Breadboard power supply module 3.3V 5V	Voltage regulator	01
MPU 6050	6DOF Accelerometer Gyroscope	01
LED	Indicator Light	01
Breadboard	-	01
Servo motor	MG995R	01
Pulley	Gear for pulling the ankle model	02
Fishing wire	Wire for pulling the ankle model	As required
Jumper wire	Connecting wire	As required

Table I shows the hardware components employed in the proposed system.

1) 1)Arduino UNO R3: This microcontroller handles sensor data and controls the various system components.

2) DHT22 (AM2302): This digital sensor monitors temperature and humidity to ensure user comfort.

3) Li-Battery (3.7V 3000mWh): A rechargeable power source that supplies energy to the system, capable of approximately 1500 charge-discharge cycles and a battery life of 11.

4) ESP32 WROOM 32: A microcontroller with built-in Wi-Fi and Bluetooth for wireless data transmission and remote monitoring.

5) Breadboard power supply module 3.3V 5V: A voltage regulator that reduces the battery voltage to match the requirements of different components.

6) MPU6050: A 6-degree-of-freedom accelerometer and gyroscope for motion and gait analysis.

7) LED: A small light indicator that shows excess humidity and temperature.

8) Servo Motor (MG995R): A motor with a stall torque of 10 kg-cm at 4.8V was used, which dynamically adjusts ankle movement to aid users during walking. Its operating voltage is 4.8-7.2V, and it can rotate 180 degrees.

9) Pulley: A gear mechanism that helps pull the ankle model through a wire system, enabling controlled movement.

10) Fishing Wire: Used to pull the structure with the help of the pulleys. This wire was chosen as it is readily available and relatively strong. The limitations of the wire were not tested through mechanical testing.

C. System Operation

1) Initialization: The sensors, Arduino Uno, and ESP32 are initialized to obtain the necessary values.

2) Motion and Gait Analysis: The MPU6050 accelerometer and gyroscope track leg movement and orientation to provide accurate data for analysis.

3) Pitch analysis: The Y-axis movement of the MPU6050 is referred to as pitch, which is used to determine forward and backwards acceleration.

4) Irregularity of movement and DHT value check: The Arduino UNO R3 analyses the data obtained from the MPU6050 and DHT22, respectively, for pitch and humidity + temperature, to decide whether to rotate the servo motor. If the threshold temperature exceeds 350°C and the relative humidity exceeds 75%, the servo motor stops rotating. This evaluation was based on the studies done by Guevara [9] , who stated a homeostatic skin temperature range of 33-36°C. Exoskeletons are expected to follow this range.

5) LED Indicator: If the DHT values exceed the threshold, an LED blinks as a warning to take off the exoskeleton and eventually stops the servo motor from rotating.

6) Data Collection and transmission: Sensor data is displayed on the Arduino serial monitor and collected in CSV format for transmission to ESP32 via serial communication. Using Wi-Fi, the ESP32 sends this data to ThingSpeakTM, a cloud-based platform, via API keys for visualization on a custom dashboard.

7) Data storage and further analysis: The data is stored on a cloud server, from which it can be accessed, downloaded in CSV format or analyzed in MATLAB. Physicians can remotely examine patients' foot health and provide treatment.

The detailed steps of the system operation are visually depicted in Fig. 2.

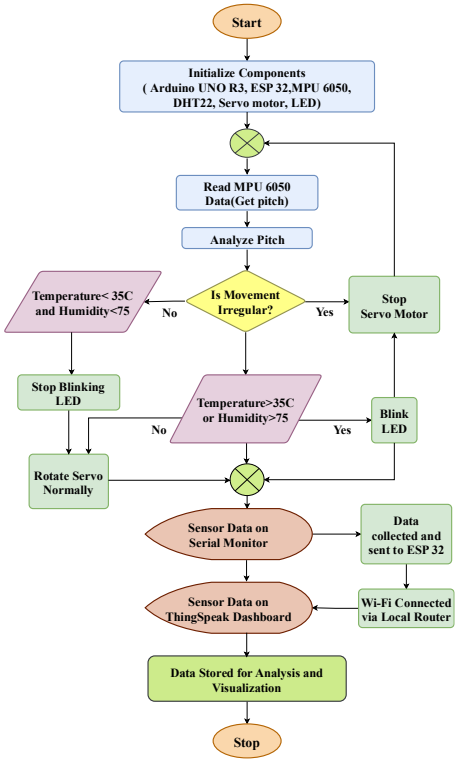


Fig. 2. Operational workflow of the IMU based exoskeleton.

IV. Implementation

A. Circuit Diagram

The circuit diagram was made using EasyEDA online circuit design and simulator [10].

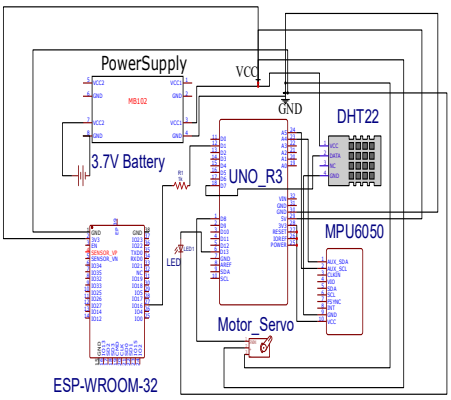


Fig. 3. Circuit diagram of the IMU based foot exoskeleton made via EasyEDA online circuit design and simulator.

B. Exoskeleton setup

To construct the prototype, low-cost materials such as spare aluminium parts, bolts, and pulleys were utilized. For comfort around the shank and foot, TPU was fabricated using a 3D printer and attached to the aluminium components with bolts. The servo motor was also mounted on top of the rod with screws and bolts, with the pulley affixed to it. To create a tendon-like pulling mechanism, high-tensile fishing wire was employed between the servo motor pulley and the foot section. Pulley attached to the ankle joint. This allowed the feet-like movement of the exoskeleton.

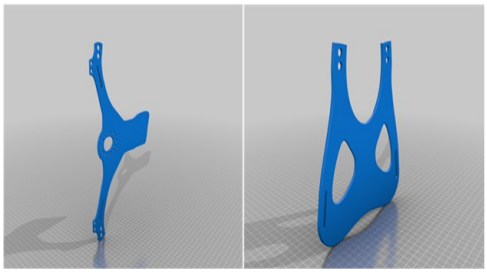


(a) (b)

Fig. 4. Footage of the structure of the exoskeleton (a) foot part, (b) shank part.

C. Open-source 3D printed parts

The 3D-printed parts were used in the shank and at the foot of the device. Thermoplastic Polyurethane (TPU) was used as it is flexible and ductile. The 3D models were downloaded from an open-source website, Thingiverse [11]. Later, it was resized in Ultimaker CURA software.



(a) (b)

Fig. 5. Structure of the exoskeleton TPU parts in Ulitimaker CURA software (a) front view, (b) side view.

D. Website Interface

The ThingSpeakTM open cloud-based server was used for visualizing and storing sensor data. It is an open-source software that allows users to communicate with internet-enabled devices. It automatically generates a write API key, which is implemented in the code section of ESP32. A private router was used to access the remote server of the ThingSpeakTM dashboard. It has several features, such as adding graphs, charts, gauge meters, and numerical displays. ThingSpeakTM has integrated support for the numerical computing software MATLAB from MathWorks, allowing website users to analyze and visualize data uploaded to the platform.

V. Results and Discussion

After the circuit was implemented, several tests were conducted to check the operation of the proposed exoskeleton. Fig. 6 shows the raw sensor data in stationary mode.

```
10:22:34.907 -> FSR1 Pressure: 687
10:22:34.907 -> FSR2 Pressure: 688
10:22:34.907 -> Irregularity: 0.06
10:22:34.907 -> Temperature (C): 24.60
10:22:34.907 -> Humidity (%): 71.70
10:22:34.907 -> --------------------
10:22:35.043 -> --- Sensor Data ---
10:22:35.043 -> MPU Angle (Servo Position): 45
10:22:35.043 -> FSR1 Pressure: 681
10:22:35.043 -> FSR2 Pressure: 687
10:22:35.043 -> Irregularity: 0.06
10:22:35.043 -> Temperature (C): 24.60
10:22:35.043 -> Humidity (%): 71.70
10:22:35.043 -> --------------------
10:22:35.125 -> --- Sensor Data ---
10:22:35.125 -> MPU Angle (Servo Position): 45
10:22:35.125 -> FSR1 Pressure: 690
10:22:35.168 -> FSR2 Pressure: 683
10:22:35.168 -> Irregularity: 0.05
10:22:35.168 -> Temperature (C)
```

Fig. 6. Raw sensor data of DHT22 and MPU6050 in Arduino IDE serial monitor.

After analyzing the pitch data and DHT22 sensor values, the servo motor rotated according to the pitch data sent to it. This was plotted on the ThingSpeakTM website to compare pitch and servo angles and analyze the DHT22 values..

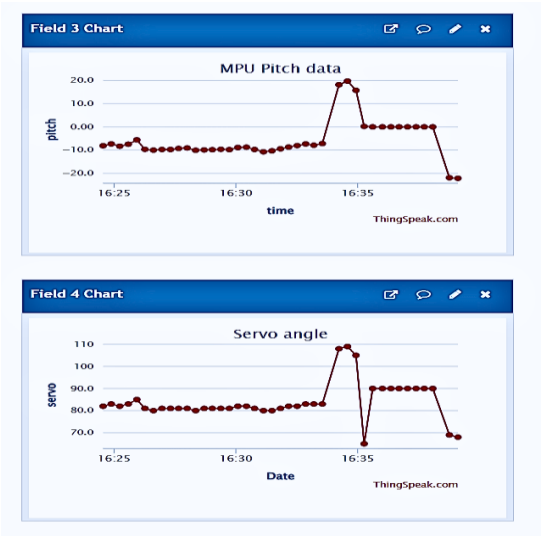


Fig. 7. Correlation between pitch and servo angle to determine the rotation of the servo motor to visualize the angle created by the ankle.



Fig. 8. ThingSpeak dashboard showing numerical and graphical data acquired from DHT22 sensor via ESP32 microcontroller via Wi-Fi.

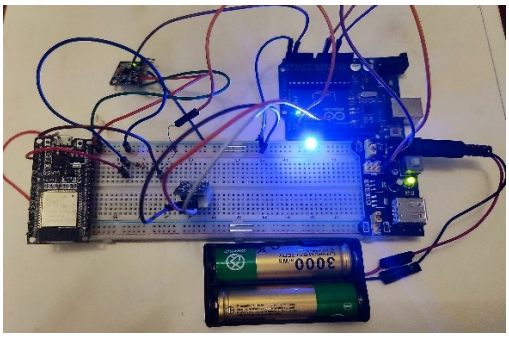


Fig. 9. Indication of excess humidity/temperature by blinking LED in the circuit of the exoskeleton.

Fig. 7 shows the pitch value obtained by MPU6050 and the servo angle generated by Arduino code simulation, demonstrating a correlation between them. For zero pitch value, the servo angle was reset to 90 degrees. Fig. 8 shows the DHT22 values being received on the dashboard. Fig. 9 shows the response to excess humidity/temperature.

There was some delay in the MPU6050 data, which may have been due to an I2C communication error or insufficient power supply. Better IMUs and better circuitry could solve this. Also, sensor data was sent with a 15-second delay on the website for delayed cloud streaming (the free version of ThingSpeak servers allows a minimum of 15-second interval between data streaming), and that is why the graphs produced in the dashboard show some latency or error. This can be overcome with the premium version of the ThingSpeak website or a customized website.

Further analysis may include plotting graphs with software such as MATLAB to visualize the servo angle. Determining the servo angle is crucial as it corresponds to the angle of the ankle joint. This was possible with ThingSpeakTM, making it a suitable website for prototype testing. All sensor data specified by fields in the website template can be exported in CSV format. Any kind of medical or lifestyle software requires CSV format data for analysis, which ThingSpeak has provided. In the future, personalized websites may be created more efficiently.

	Timestamp	Obs No	Temp	Humidity	Pitch	Servo angle
139	2025-04-22T00:33:44+06:00	138	31.2	78.5	-8.12	82
140	2025-04-22T00:34:04+06:00	139	31.2	78.5	-7.34	83
141	2025-04-22T00:34:25+06:00	140	31.2	78.4	-8.32	82
142	2025-04-22T00:34:46+06:00	141	31.2	78.6	-7.47	83
143	2025-04-22T00:35:07+06:00	142	31.2	78.6	-5.6	85
144	2025-04-22T00:35:27+06:00	143	31.2	78.4	-9.66	81
145	2025-04-22T00:35:48+06:00	144	31.2	78.5	-10	80
146	2025-04-22T00:36:09+06:00	145	31.2	78.6	-9.73	81
147	2025-04-22T00:36:29+06:00	146	31.2	78.5	-9.74	81
148	2025-04-22T00:36:50+06:00	147	31.2	78.5	-9.29	81
149	2025-04-22T00:37:11+06:00	148	31.2	78.6	-9.07	81
150	2025-04-22T00:37:31+06:00	149	31.2	78.5	-10.07	80
151	2025-04-22T00:37:52+06:00	150	31.2	78.5	-9.91	81
152	2025-04-22T00:38:13+06:00	151	31.2	78.6	-9.83	81
153	2025-04-22T00:38:34+06:00	152	31.2	78.8	-9.65	81
154	2025-04-22T00:38:55+06:00	153	31.2	78.8	-9.79	81
155	2025-04-22T00:39:15+06:00	154	31.2	78.7	-8.86	82
156	2025-04-22T00:39:36+06:00	155	31.2	78.6	-8.75	82
157	2025-04-22T00:39:57+06:00	156	31.2	78.5	-9.73	81
158	2025-04-22T00:40:18+06:00	157	31.2	78.6	-10.72	80

feeds new2

Fig. 10. . Exported CSV format data in a Microsoft Excel file from ThingSpeak dashboard for further analysis.

Fig. 10 shows the sensor values in CSV format, making it easy for further analysis, as CSV format files can be easily imported into any application, such as MATLAB.

TABLE II. OBSERVATION OF THE PITCH AND SERVO ANGLE FOR CALCULATING THE ERROR OF THE EXOSKELETON MOVEMENT.

Obs. no	Pitch (°)	Received Servo (°)	Expected Servo (90°+pitch) (°)	Comment
01	-9.73	81	80.27	Near perfect
02	-10.72	80	79.28	
03	-10.32	80	79.68	
04	-9.44	81	80.56	
05	-8.71	82	81.29	
06	-8.05	82	81.95	
07	-7.34	83	82.66	
08	-7.9	83	82.6	
09	-7.18	83	82.82	
10	18.16	108	108.16	
11	19.74	109	109.74	
12	15.76	105	105.76	
13	0.21	65	90.21	Spike value
14	0	90	90	Sensor off
15	0	90	90	Sensor off
16	0	90	90	Sensor off
17	0	90	90	Sensor off
18	9.73	81	80.27	Restart
19	10.72	80	79.28	
20	10.32	80	79.68	

Table II presents a comparison between pitch data servo angles and calculates their error. IMU motion was manually simulated to observe servo response.

Anomalies were observed, such as in observation 13, where the pitch data suddenly returned to zero value. It may have been due to a power failure or an I2C communication error. A sudden power draw, interference, or the use of long wires may cause corruption in I2C lines, which can freeze the input of MPU6050. Sometimes, servo motors generate electromagnetic noise that is transmitted back to the microcontroller, causing spikes or delays. This issue may be resolved by using more reliable power sources with sufficient amperage or using decoupling capacitors. Observations 14 to 17 record zero value of pitch, suggesting that the MPU6050 may have temporarily powered off or lost communication.

TABLE III. SELECTED OBSERVATION OF RECEIVED SERVO ANGLE AND NORMAL ANKLE ANGLE TO CALCULATE STANDARD DEVIATION.

%gait	Pitch (°)	Received servo angle (°)	Normal ankle angle (°)	Normal servo (°)
0	0.00	90	0	90
10	-9.73	81	-10	80
20	-9.44	81	-10	80
30	-7.18	83	-3	87
40	-7.90	83	+3	93
50	18.16	108	+10	100
60	19.74	109	+15	105
70	9.73	81	+8	98
80	-8.71	82	0	90
90	-10.72	80	-5	85

Table III represents the correlation between hand-rotated IMU tests and standard human gait profiles. Here, a normal ankle angle was taken from Klaewkasikum et al. [12]. Normal servo angle represents the expected servo angle according to normal ankle angle. By comparing these data, the prototype actuation was compared to real human gait. The following equations were used for error analysis:

$$\text{Error, } e_i = Received_Servo_i - (90 \circ + Pitch_i) \quad (1)$$

$$\text{Mean error, } \bar{e} = \frac{1}{N}\sum_{i=1}^{N} e_i \quad (2)$$

$$\text{Standard deviation, (SD) } \sigma = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(e_i - \bar{e})^2} \quad (3)$$

$$SD = 4.86°$$

McGinley et al. [13] showed gait analysis standards treat errors <5° as acceptable. Although an $SD < 2°$ is considered very good, this prototype showed some inefficiency due to off-the-shelf components and power sources.

The results presented in this section confirm the functionality of this project and demonstrate that physicians can remotely download patients' data and analyze it for further treatment. Pre-built MATLAB, GUI's or custom dashboards may be created for their comfortability. By automating the insertion of CSV data into analysis software, personalized treatment can be provided.

However, this project was designed for demonstration purposes only and was not tested on human subjects or worn on the human body. Future improvements to this project would require mechanical and electrical safety assessments, especially regarding skin contact temperature and actuation control.

VI. CONCLUSION

This project marks the first steps in developing a low-cost smart exoskeleton. As Bangladesh is a developing country, high-priced exoskeletons are out of reach of the general public, and post-diabetic syndromes such as foot drop are becoming frequent. This project successfully collects and transmits sensor data, such as that from the MPU6050 and DHT22 integrated on Arduino UNO and ESP32, analyzes it, and simulates real ankle movement using a tendon-like mechanism with real-time visualization of sensor data on ThingSpeakTM. For this reason, the exoskeleton was designed to target ankle movement only. This project distinguishes itself from other low-cost exoskeletons by generating actuation from an IMU sensor. Control algorithms were integrated to rotate the motor according to the pitch created by the sensor. Made from spare aluminium parts and 3D-printed TPU, it proved to be a low-cost solution. The MPU6050 sent pitch data, which generated servo angles to rotate the servo motor, causing the exoskeleton to move in the sagittal plane. The feature that distinguishes this exoskeleton from others is its integrated IoT system, which transmits sensor data that can be later analyzed by physicians for remote healthcare, representing the main attraction of this project. The results and analysis of this project validate this and demonstrate that doctors can remotely access and analyze patient data to support ongoing treatment.

However, the MPU occasionally returned zero values, which may have been due to an I2C communication error or insufficient power, resulting in improper rotation of the servo motor. Another issue was the streaming rate of ThingSpeakTM, which was limited to a 15-second time interval (free version). This led to the discontinuation of data flow. Although the prototype was made using affordable materials, it wasn't strong enough to be worn safely. The servo motor lacked torque.

Future improvements involve finding comparatively low-cost yet strong materials for manufacturing the structure and creating a personalized website rather than using ThingSpeakTM for real-time data streaming and powerful

machine learning algorithms for analyzing gait patterns. Powerful motors would be used for better torque and strength. To avoid complex circuitry, the circuit can be implemented on a printed circuit board (PCB). Despite these challenges, this prototype lays the groundwork for a smart, wearable, and affordable assistive exoskeleton.

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